

Arnav Khairari

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EDUCATION

Rutgers University, New Brunswick (Dean's Lists) (GPA:3.7)

May 2027

Data Science B.S & Computer Science B.S

EXPERIENCE

Tenneco – Data Science Intern

Sep 2026-Present

- Supporting analytics team, building **predictive models** in **Python** for SKU-level inventory and **demand-forecasting** decisions.

Merck & Co. – Global Clinical Data Standards (GCDS) Intern

June 2026-Aug 2026

- Engineered a **scalable Python/Dataverse pipeline** via paginated OData REST APIs, transforming **101K+ records across 743 clinical studies** into real-time workload, capacity, and milestone-risk metrics with multi-layer caching to prevent data bottlenecks.
- Applied **exploratory data analysis** and **feature engineering** to **unlabeled operational data**, reconstructing training labels and engineering workload/study features to build a supervised steward-prediction model achieving an 84.6% match rate for allocation.
- Evolved a **Microsoft Copilot Studio prototype** into a scalable **Python/Streamlit RAG + ML** with embeddings-based retrieval, evaluating retrieval quality and response grounding to replace manual workflows with reliable, real-time clinical insights.

Tenneco – Supply Chain Data Analytics Intern

May 2025-May 2026

- Built **end-to-end SKU analytics models** in **Python** and **Excel** that quantified SKU-level contribution, eliminating planning bottlenecks, improving demand-risk forecasting accuracy by 20%, **tripling decision-making speed** through automated insights.
- Designed a **Beck Arnley analytics portal** using **Power BI** and **Tableau**; applied classification and segmentation to highlight high-impact SKUs across business lines, enabling 10+ planners to run **targeted inventory strategies**, sustaining 95% fill rate.
- Developed and automated **weekly fill-rate dashboards** in **Power BI** with **SAP** integrations, optimizing **ETL pipelines and M-code**; improved scalability and leadership visibility across 15+ product lines to support **predictive inventory planning**.

Cerebras Systems AI - Developer Fellow

Oct 2024- May 2025

- Constructed a Python-based multi-agent system for task automation on accelerated compute infrastructure, improving task allocation efficiency by 23% using LangChain/LangGraph, containerized (Docker) for scalable, parallelized deployment.
- Adopted a **Flask** framework for a task management web application, resulting in the ability to handle over **83 tasks** simultaneously while integrating generative AI features that classified and generated responses based on user interactions.

PROJECTS

Gesture-Controlled Interface System — Personal Project

Sep 2026

- Architected a **real-time computer vision** and **gesture recognition pipeline** in Python using **MediaPipe**, applying signal smoothing and a finite-state machine with temporal debouncing to suppress false-positive triggers across gesture classes.
- Decoupled detection from action execution via a **modular strategy-pattern dispatch layer** with per-user calibration for human-machine interaction, enabling touchless, contamination-safe control across sterile, cleanroom, and environments.

Modeling & Analyzing Tablet Characteristics — Pharmaceutical Manufacturing Project

Dec 2025

- Developed **predictive models** in **R** (**glmnet**, **tidyverse**) using **ANOVA** and **cross-validated LASSO** to quantify how continuous manufacturing parameters impact tablet hardness and density; achieved $R^2 \approx 0.84$ (SFAPAP) and **0.72** (dAPAP).
- Generated interpretable insights for **pharmaceutical process optimization and scale-up**, identifying main compression height as the primary critical process parameter (CPP) influencing tablet quality attributes.

NeuroDrive Helper – HackRU Project

Oct 2025

- Built a Chrome extension with a FastAPI backend integrating two pretrained **NLP models** from Hugging Face for text **simplification and summarization**; added **ElevenLabs TTS** for natural speech, **containerized with Docker** for <2s latency.
- Designed and deployed **5+ accessibility features**—including dyslexia-friendly fonts, color-blind filters, high-contrast themes, distraction-free overlays, and text-to-speech—expanding usability for individuals with visual and cognitive challenges

Supervised Learning Models

Jan 2024- May 2024

- Engineered regression tree models on **7,500+ student records** (GPA, major, connections) to estimate earnings; applied hyperparameter tuning with k-fold cross-validation, reducing MSE (Mean Squared Error) by **~65%**.
- Programmed features (age brackets, symptom indicators) and trained **Naive Bayes & CART classifiers on 400+ patient profiles**; achieved up to **92% accuracy** (CART), validated via confusion matrices

TECHNICAL SKILLS

- Languages:** Python, R, Java, SQL, C, JavaScript, HTML, MATLAB
- Data Science & AI:** Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, caret, rpart, LangChain, LangGraph, RAG, Streamlit
- Tools & Platforms:** Power BI, OData REST APIs, SAP, AWS, Azure, Firebase, Docker, Git, React.js, Next.js, Flask, Tabular
- Coursework:** Multivariable Calculus, Linear Optimization, Regression Method, Machine Learning Principle